

Building a Vietnamese Secondary Informatics AI Tutor Benchmark from Teacher-Authored Dialogues

Author names pending confirmation

Affiliations pending confirmation

Vietnam

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I. INTRODUCTION

Large language models (LLMs) are increasingly capable of producing correct answers and fluent explanations, but these abilities do not by themselves establish that a model can tutor effectively. Tutoring also requires (i) interpreting what a learner currently understands; (ii) diagnosing errors when evidence permits; (iii) selecting an appropriate pedagogical move; (iv) calibrating the amount of support; and (v) communicating in a way that preserves the learner’s opportunity to think. Recent tutoring benchmarks support this distinction. MathTutorBench reports that mathematical problem-solving ability does not directly translate into pedagogical quality [1]; KMP-Bench separates verifiable mathematical skills from principle-guided dialogue quality [2]; and TutorBench evaluates responses with fine-grained, sample-specific criteria [3]. Evaluating a tutor is therefore an assessment-design problem rather than only a question-answering problem.

Prior AI-tutoring benchmarks leave three gaps for our target setting. First, their domain coverage is concentrated in mathematics or broad STEM curricula, while specialized school subjects remain less explored. Second, they operate predominantly in English, leaving language- and curriculum-specific tutoring behavior underrepresented. Third, data construction is not unified: (i) KMP-Bench builds pedagogically planned dialogue flows; (ii) TutorBench uses expert-authored situations and rubrics; and (iii) MathTutorBench assembles existing tutoring datasets under predefined tasks and metrics. In each case, prior assumptions about tasks or pedagogical situations structure the data before evaluation. Less is known about starting from ordinary, teacher-authored pedagogical dialogues collected independently of a benchmark taxonomy and converting them into traceable evaluation units.

We address these gaps by constructing a Vietnamese lower-secondary Informatics AI-tutor benchmark for Grades 6–9. Rather than generating conversations from predefined benchmark situations, our process (i) begins with raw teacher-authored dialogues; (ii) applies auditable quality control; and (iii) converts eligible tutor turns into next-response evaluation candidates. The benchmark then connects six pedagogical

principles with six tutor capabilities to specify what a response should accomplish and how its quality should be evaluated.

The paper makes three contributions:

- 1) a traceable construction pipeline that transforms 1,050 raw teacher-authored dialogues into 665 quality-controlled dialogue families and 2,028 turn-level benchmark candidates;
- 2) a measurement specification that combines six pedagogical principles with six tutor capabilities through response requirements, a common rubric tier, and principle-specific rubric tiers; and
- 3) an analysis of pedagogical-requirement coverage and benchmark readiness, identifying 1,400 candidates ready for downstream construction and 628 candidates routed to targeted review.

Together, these contributions establish a curriculum-grounded and traceable basis for evaluating AI tutoring responses in Vietnamese lower-secondary Informatics.

II. RELATED WORK AND BACKGROUND

A. Pedagogical and Measurement Foundations

The pedagogical vocabulary in this work begins with Alison and Tharby’s six principles: (i) Challenge; (ii) Explanation; (iii) Modelling; (iv) Practice; (v) Feedback; and (vi) Questioning [4]. These principles are interrelated elements of teaching rather than mutually exclusive task classes. They provide a functional description of what a tutor response may need to accomplish: (i) sustaining appropriate cognitive effort; (ii) clarifying knowledge; (iii) demonstrating application; (iv) creating opportunities to practise; (v) using learner performance to guide improvement; and (vi) eliciting evidence of understanding.

Our capability model also draws on research on adaptive scaffolding. Van de Pol *et al.* distinguish the means of support—including questioning; explaining; modelling; and feedback—from four properties that make support contingent on the learner: (i) diagnosis of current competence; (ii) calibrated assistance; (iii) fading; and (iv) transfer of responsibility [5]. This distinction motivates separating (i) pedagogical-strategy selection from scaffolding calibration; and (ii) observation of a learner’s current state from diagnosis of the cause of an error or impasse.

Evidence-centered design further requires assessment developers to link claims about latent capabilities to tasks and observable evidence [6]. Contemporary validity theory similarly

treats content review as one component of a broader argument about the intended interpretation and use of assessment results [7]. Following these foundations, we organize tutor quality into six capabilities: (i) subject-matter accuracy and grounding; (ii) learner-state modelling; (iii) pedagogical-strategy selection; (iv) scaffolding calibration; (v) diagnosis; and (vi) clear and supportive communication appropriate to the learner’s age. The principles specify response-level pedagogical requirements, whereas the capabilities define the qualities observed by the rubric.

B. Benchmarks for LLM Tutoring

Existing tutoring benchmarks operationalize tutoring quality in substantially different ways. MathTutorBench organizes seven tasks under three groups: (i) subject expertise; (ii) student understanding; and (iii) pedagogical response generation [1]. Structured outputs use task-specific metrics, whereas open-ended tutor responses are evaluated with a learned preference model. This makes the solver–tutor distinction explicit and avoids reducing all tutoring ability to answer accuracy.

KMP-Bench provides the closest structural precedent for our turn-level candidates [2]. Its KMP-Dialogue module truncates a dialogue before a tutor turn and associates each planned tutor action with one or two of the six pedagogical principles. A candidate and a reference response are compared using four general criteria plus three criteria for every assigned principle, with one of three criterion-level judgments: (i) Win; (ii) Tie; or (iii) Lose. Its separate KMP-Skills module uses different, task-appropriate metrics for (i) problem solving; (ii) error handling; and (iii) problem generation. The principles therefore select relevant aspects of dialogue quality rather than act as mutually exclusive benchmark tasks.

TutorBench instead contains 1,490 expert-curated high-school and Advanced Placement STEM samples covering (i) adaptive explanation; (ii) assessment and feedback; and (iii) active-learning support [3]. Each sample has its own atomic criteria, weighted by the importance and direction of the requirement, and an LLM judge evaluates each criterion independently. This provides diagnostic granularity but requires a large sample-specific rubric-authoring effort.

These designs establish three lessons for our benchmark: (i) correctness and tutoring quality must be measured separately; (ii) a reference response is a comparator rather than the only valid wording; and (iii) four constructs play distinct roles: tasks; pedagogical principles; tutor capabilities; and rubrics. We therefore (i) use one next-response generation task; (ii) represent pedagogical demand through per-candidate principle requirements; and (iii) use capabilities to organize observable scoring criteria.

C. Vietnamese and Domain-Specific Educational Evaluation

Vietnamese LLM evaluation has expanded beyond translated general-purpose tests. VNHSGE collects more than 19,000 multiple-choice questions and 300 literary essays from Vietnamese high-school examinations across nine subjects [8]. ViMath-Bench focuses on Vietnamese mathematical reasoning

[9], while VMLU evaluates four areas: (i) general knowledge; (ii) reading comprehension; (iii) reasoning; and (iv) general multi-turn conversation. Its subject inventory includes applied Informatics and several university-level computing areas [10]. These resources establish language- and curriculum-specific evaluation, but their primary units are examination answers, reasoning outputs, or general conversations rather than context-conditioned tutor responses.

Vietnamese pedagogical evaluation has also begun to emerge. Outside the peer-reviewed literature, the publicly announced DeepEduBench initiative reports 13,700 Vietnamese items used to assess 52 models on (i) stepwise explanation; (ii) learner adaptation; (iii) example and exercise selection; and (iv) focused feedback [11]. We use this source only to document the initiative’s existence and stated scope, not to infer its methodology. A separate workshop paper evaluates LLMs as tutors for Vietnamese K–12 learners using 540 yes/no questions in (i) mathematics; (ii) physics; and (iii) chemistry [12]. The available descriptions do not report a benchmark slice for the national lower-secondary Informatics curriculum, candidates reconstructed from ordinary teacher-authored dialogues, or a next-response protocol combining pedagogical principles with tutor capabilities.

The domain-specific literature closes a different part of the gap. CSTutorBench evaluates 11 models as tutors in a middle-school block-programming and robotics environment through 17 scenario-based questions and an eight-criterion pedagogical rubric [13]. Shin *et al.* instead analyze 249 authentic CodeHelp exchanges from an undergraduate C++ course using criteria for (i) mistake identification; (ii) mistake location; (iii) guidance; (iv) accuracy; and (v) next steps [14]. Multilingual programming work has also generated exercises in Vietnamese, but does not evaluate dialogue-level tutoring quality [15]. Thus, prior work covers Vietnamese pedagogical evaluation and computing-specific tutor evaluation separately. Our literature search found no publicly documented benchmark that combines (i) Vietnamese response generation; (ii) the Grades 6–9 Informatics curriculum; (iii) conversation-conditioned next-turn tutoring; and (iv) candidates derived from traceable teacher-authored pedagogical dialogues. We therefore position this work at that specific intersection.

III. DATASET AND CONSTRUCTION

To be completed.

IV. PEDAGOGICAL REQUIREMENTS AND RUBRIC DESIGN

To be completed.

V. INITIAL ANALYSIS

To be completed.

VI. DISCUSSION AND LIMITATIONS

To be completed.

VII. CONCLUSION

To be completed.

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